

Sheet 3

TA: Viktoriia Pinchenkova, viktoriia.pinchenkova@unibas.ch, office 4.6

To be discussed on Tuesday, October 6, 16:00

Submission deadline: Monday, October 5, 23:59 (Basel time)

Please write your full name and student ID number directly on the assignment.

Problems 1–6 are required. Problem 7 is optional. An attempt at Problem 7 will be taken into account when assessing your effort, but there is no penalty for leaving it out.

Conventions. All matrices are written in the ordered computational basis unless stated otherwise. The symbol $|0\rangle$ denotes a basis vector, not the zero vector $\mathbf{0}$.

1. Computational Basis and Vector Spaces

Work in $\mathbb{C}^3$ with the orthonormal computational basis $|0\rangle = (1, 0, 0)^T$, $|1\rangle = (0, 1, 0)^T$ and $|2\rangle = (0, 0, 1)^T$.

- Write the matrices $|0\rangle\langle 0|$, $|0\rangle\langle 1|$ and $|2\rangle\langle 0|$. Give a general rule for the matrix entries of $|i\rangle\langle j|$, where $i, j \in \{0, 1, 2\}$; you do not need to write the other six matrices.
- Evaluate $\sum_{i=0}^2 |i\rangle\langle i|$.
- Which of the following sets are vector subspaces of $\mathbb{C}^3$? Justify your answers:

$$W = \{(0, z_1, z_2)^T : z_1, z_2 \in \mathbb{C}\}, \quad U = \{|v\rangle \in \mathbb{C}^3 : \langle v|v\rangle = 1\}.$$

2. Inner Products and Norms

Let $|v\rangle = (1, i)^T$ and $|w\rangle = (2, 1 - i)^T$ in $\mathbb{C}^2$.

- Write $\langle v|$ and $\langle w|$, compute $\langle v|w\rangle$ and $\langle w|v\rangle$, and verify the conjugate-symmetry relation $\langle w|v\rangle = \langle v|w\rangle^*$.
- Find the norms of both vectors and write the normalised vectors $|\hat{v}\rangle$ and $|\hat{w}\rangle$ explicitly.

3. Bra–Ket Algebra

Let $|u\rangle, |v\rangle, |w\rangle \in \mathbb{C}^n$ be unit vectors and let $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear operator. The vectors are not assumed to be mutually orthogonal. Simplify each expression and state whether its result is a scalar, a ket vector or an operator:

- $(|u\rangle\langle v|)(|v\rangle\langle w|)$;
- $(\langle u|A|v\rangle)|w\rangle$;
- $(|u\rangle\langle u|)^2$;
- $\langle w|(|u\rangle\langle v|)|w\rangle$.

4. Qubits, Measurement and Antipodal States

A pure qubit state, up to a global phase, can be written as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi.$$

Its Bloch vector is $\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$. Recall the Born rule: in an orthonormal measurement basis $\{|a_j\rangle\}$, the probability of outcome j is $|\langle a_j | \psi \rangle|^2$.

- (a) Starting from two complex amplitudes $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, explain why a pure qubit state has two independent real parameters. Compute the computational-basis measurement probabilities. Show that multiplying $|\psi\rangle$ by $e^{i\gamma}$, with $\gamma \in \mathbb{R}$, changes no measurement probability in any orthonormal basis.
- (b) Using Dirac notation, show that antipodal Bloch vectors $\mathbf{r}$ and $-\mathbf{r}$ correspond to orthogonal states.
- (c) Let $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Compare their measurement probabilities in the computational basis and in the basis $\{|+\rangle, |-\rangle\}$.

5. Bloch-Sphere Coordinates

Use the parametrisation and angle ranges in Problem 4. The Pauli matrices are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (a) Find (θ, ϕ) and the Bloch vector for each of the states

$$|0\rangle, \quad |1\rangle, \quad |+\rangle, \quad |-\rangle, \quad \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

- (b) Find a normalised ket, in the standard Bloch form, corresponding to $\mathbf{r} = (1, 1, 1)/\sqrt{3}$. Give its amplitudes exactly, not only as decimal approximations.
- (c) By explicit matrix multiplication, show that for a normalised pure state $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$,

$$(\langle\psi|X|\psi\rangle, \langle\psi|Y|\psi\rangle, \langle\psi|Z|\psi\rangle) = (2\operatorname{Re}(\alpha^*\beta), 2\operatorname{Im}(\alpha^*\beta), |\alpha|^2 - |\beta|^2).$$

Substitute the Bloch parametrisation and show that this is precisely $\mathbf{r}$.

6. Linear Operators and Their Matrix Representations

Let $B : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear operator satisfying $B|j\rangle = |\psi_j\rangle$ for $j = 0, \dots, n-1$. The vectors $|\psi_j\rangle$ need not be normalised or linearly independent.

- (a) Express B as a sum of outer products. Explain why its action on a basis determines its action on every vector.
- (b) Show that the coordinate column of $|\psi_j\rangle$ is column j of the matrix of B .

(c) For the Hadamard operator

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

find $H|0\rangle$, $H|1\rangle$ and an outer-product expression for H . Use linearity to find $H(\alpha|0\rangle + \beta|1\rangle)$.

7. Mixed States and the Bloch Ball (Optional)

Background for this optional problem. If a preparation produces normalised states $|\psi_k\rangle$ with classical probabilities $q_k \geq 0$, $\sum_k q_k = 1$, its density matrix is

$$\rho = \sum_k q_k |\psi_k\rangle\langle\psi_k|.$$

The probability of an outcome associated with a normalised vector $|a\rangle$ in an orthonormal measurement basis is

$$p(a) = \langle a|\rho|a\rangle = \sum_k q_k |\langle a|\psi_k\rangle|^2.$$

A qubit density matrix has the Bloch representation

$$\rho = \frac{1}{2}(I_2 + r_x X + r_y Y + r_z Z).$$

Here $\mathbf{r} \in \mathbb{R}^3$, and physical states fill the Bloch ball $\|\mathbf{r}\| \leq 1$; pure states lie on its surface.

- (a) A preparation produces $|+\rangle$ with probability p and $|-\rangle$ with probability $1-p$, where $0 \leq p \leq 1$. Write its density matrix explicitly, find its Bloch vector and locate it in the Bloch ball. For which values of p is the state pure?
- (b) At $p = 1/2$, compare this mixture with the coherent superposition $|\chi\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ by computing the computational-basis measurement probabilities for both preparations.
- (c) Why can a vector $\mathbf{r}$ with $\|\mathbf{r}\| > 1$ not represent a physical qubit state? Consider the matrix $\rho_{\mathbf{r}} = (I_2 + r_x X + r_y Y + r_z Z)/2$. *Hint:* Choose a pure state $|a\rangle$ with Bloch vector $-\mathbf{r}/\|\mathbf{r}\|$, and use Problem 5(c) to compute $\langle a|\rho_{\mathbf{r}}|a\rangle$.